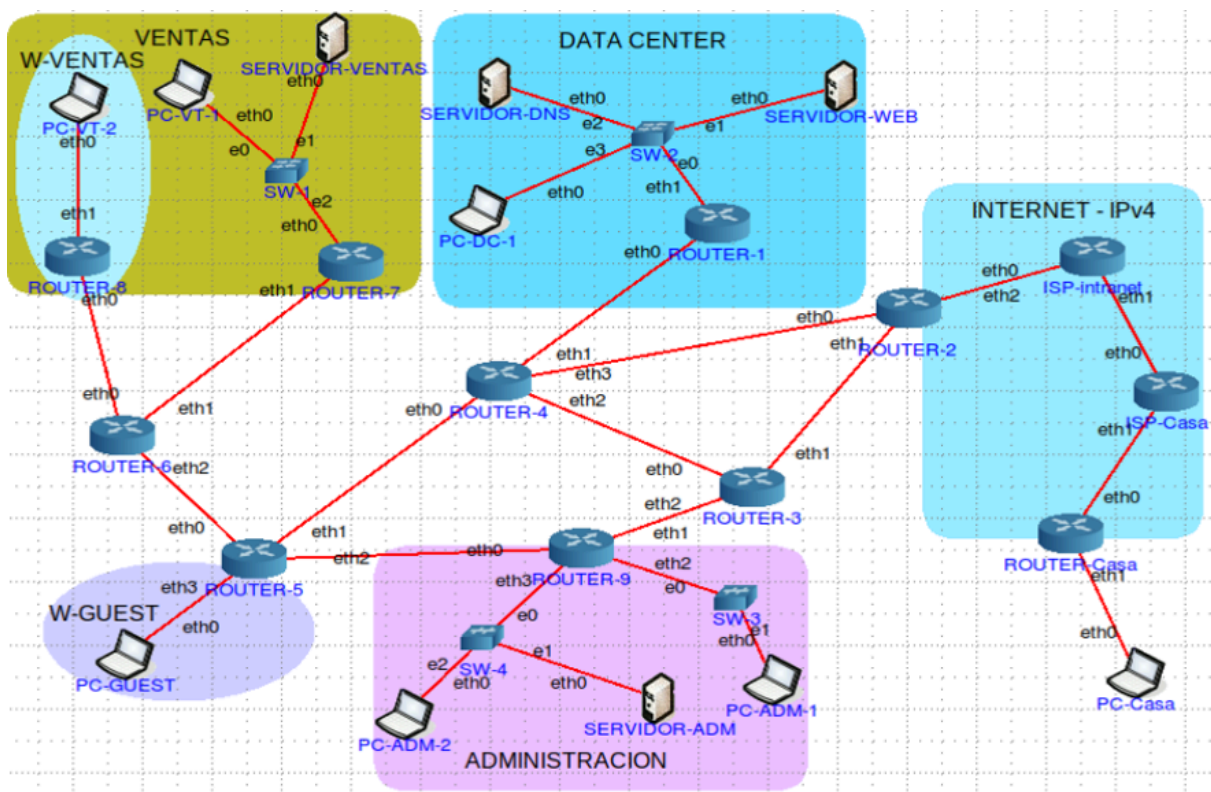


Características de la red

- Todos los equipos (incluye hosts y routers) pueden comunicarse entre sí dentro de la intranet.
- VENTAS puede comunicarse con otros equipos en la Internet
- El SERVIDOR-WEB puede ser accedido por cualquier equipo de la Internet, en la dirección (global) provista por el DNS (servidorweb.abc.com).
- El SERVIDOR-DNS tiene una dirección global que permite que sea accedido desde afuera de la intranet.
- Desde la red inalámbrica de W-GUEST no es posible acceder a equipos internos de la intranet, excepto al SERVIDOR-DNS y al SERVIDOR-WEB por sus direcciones globales.
- Los paquetes que se procesan en el ROUTER-5 originados en W-GUEST solo pasan por el ROUTER-9, mientras que el resto del tráfico procesado en el ROUTER-5 se direcciona al ROUTER-4.
- Los paquetes TCP con destino fuera de la intranet pasan por el Router 3, mientras que el resto del tráfico dirigido fuera de la intranet no pasan por dicho equipo.
- Se puede acceder a cualquiera de los routers desde fuera de la intranet.
- Todas las interfaces de INTERNET sólo tienen direcciones IPv4. El resto de las interfaces sólo tienen direcciones IPv6.



Diseño de Topología y Direccionamiento

Asignación de Subredes (IPv6 Global)

El proveedor asignó el prefijo global 2001:1200:0:21f0::/60. Este fue subdividido en múltiples redes /64 para cada segmento interno y enlaces punto a punto /127 entre routers. Todos los routers y hosts finales recibieron direcciones globales

Nombre de Red	Dirección/Prefijo
Data Center	2001:1200:0:21f7::/64
W-Ventas	2001:1200:0:21f3::/64
Ventas-1	2001:1200:0:21f2::/64
W-Guest	2001:1200:0:21f4::/64
ADM-1	2001:1200:0:21f5::/64
ADM-2	2001:1200:0:21f6::/64
R5 (Router 2- Router 4)	2001:1200:0:21f0::0/127
R6 (Router 3- Router 2)	2001:1200:0:21f0::2/127
R7 (Router 4- Router 1)	2001:1200:0:21f0::4/127
R8 (Router 4- Router 5)	2001:1200:0:21f0::6/127
R9 (Router 4- Router 3)	2001:1200:0:21f0::8/127
R10 (Router 5- Router 9)	2001:1200:0:21f0::a/127
R11 (Router 9- Router 3)	2001:1200:0:21f0::c/127
R12 (Router 6- Router 5)	2001:1200:0:21f0::e/127
R13 (Router 7- Router 6)	2001:1200:0:21f0::10/127
R14 (Router 8- Router 6)	2001:1200:0:21f0::12/127

Nombre de equipo	Dirección/Prefijo global
Pc-Casa	2001:1200:0:21f1::1/64
Router Casa - eth 0	192.168.1.1/64
Router Casa - eth 1	2001:1200:0:21f1::/64
ISP Casa - eth 0	192.168.2.1/64
ISP Casa - eth 1	192.168.1.2/64
ISP Intranet - eth 0	192.168.3.1/64
ISP Intranet - eth 1	192.168.2.2/64
Router 2 - eth 0	2001:1200:0:21f0::0/127
Router 2 - eth 1	2001:1200:0:21f0::2/127
Router 3 - eth 0	2001:1200:0:21f0::8/127
Router 3 - eth 1	2001:1200:0:21f0::3/127
Router 3 - eth 2	2001:1200:0:21f0::c/127
Router 4 - eth 0	2001:1200:0:21f0::6/127
Router 4 - eth 1	2001:1200:0:21f0::5/127
Router 4 - eth 2	2001:1200:0:21f0::9/127
Router 4 - eth 3	2001:1200:0:21f0::1/127
Router 9 - eth 0	2001:1200:0:21f0::b/127
Router 9 - eth 1	2001:1200:0:21f0::d/127
Router 9 - eth 2	2001:1200:0:21f5::/64
Router 9 - eth 3	2001:1200:0:21f6::/64
Pc-Adm-1	-
Pc-Adm-2	-
Servidor Adm	-
Router 5 - eth 3	2001:1200:0:21f4::/64
Pc-Guest	2001:1200:0:21f4::1/64
Router 6 - eth 0	2001:1200:0:21f0::12/127
Router 6 - eth 1	2001:1200:0:21f0::10/127
Router 6 - eth 2	2001:1200:0:21f0::f/127
Router 8 - eth 0	2001:1200:0:21f0::13/127
Router 8 - eth 1	2001:1200:0:21f3::/64
Pc-Vt-2	2001:1200:0:21f3::1/64

Router 7 - eth 0	2001:1200:0:21f2::/64
Router 7 - eth 1	2001:1200:0:21f0::11/127
Pc-Vt-1	2001:1200:0:21f2::2/64
Servidor Ventas	2001:1200:0:21f2::1/64
Router 1 - eth 0	2001:1200:0:21f0::4/127
Router 1 - eth 1	2001:1200:0:21f7::/64
Pc-Dc-1	2001:1200:0:21f7::3/64
Servidor DNS	2001:1200:0:21f7::2/64
Servidor WEB	2001:1200:0:21f7::1/64

Direcciones Únicas Locales (ULA)

Para facilitar la administración interna y reducir la dependencia de direcciones globales, se implementó un esquema ULA en el espacio fd00::/8, proporcionando direccionamiento estable para todos los dispositivos internos.

Red/Dispositivo	Prefijo/Dirección ULA
Servidor WEB	fd00:0:0:7::1/64
Servidor DNS	fd00:0:0:7::2/64
Pc-Dc-1	fd00:0:0:7::3/64
Router 1 - eth 2	fd00:0:0:7::/64
Servidor Ventas	fd00:0:0:2::1/64
Pc-Vt-1	fd00:0:0:2::2/64
Router 7 - eth 0	fd00:0:0:2::/64
Pc-Vt-2	fd00:0:0:3::1/64
Router 8 - eth 1	fd00:0:0:3::/64
Router 5 - eth 3	fd00:0:0:4::/64
Router 9 - eth 3	fd00:0:0:6::/64
Router 9 - eth 2	fd00:0:0:5::/64
Pc-Adm-2	fd00:0:0:6::2/64
Servidor Adm	fd00:0:0:6::1/64
Pc-Adm-1	fd00:0:0:5::1/64
Pc-Casa	fd00:0:0:1::1/64

Router Casa	fd00:0:0:1::/64
Router 2 - eth 0	fd00::/127
Router 2 - eth 1	fd00::2/127
Router 3 - eth 0	fd00::9/127
Router 3 - eth 1	fd00::3/127
Router 3 - eth 2	fd00::c/127
Router 4 - eth 0	fd00::6/127
Router 4 - eth 1	fd00::5/127
Router 4 - eth 2	fd00::9/127
Router 4 - eth 3	fd00::1/127
Router 9 - eth 0	fd00::b/127
Router 9 - eth 1	fd00::d/127
Router 6 - eth 0	fd00::12/127
Router 6 - eth 1	fd00::10/127
Router 6 - eth 2	fd00::f/127
Router 8 - eth 0	fd00::13/127
Router 7 - eth 1	fd00::11/127
Router 1 - eth 0	fd00::4/127

Las direcciones de los equipos se configuraron manualmente con los siguiente comandos:

Servidor WEB:

- ip -6 addr add fd00:0:0:7::1/64 dev eth 0
- ip -6 addr add 2001:1200:0:21f7::1/64 dev eth0

Servidor DNS:

- ip -6 addr add fd00:0:0:7::2/64 dev eth 0
- ip -6 addr add 2001:1200:0:21f7::2/64 dev eth0

Pc-Dc-1:

- ip -6 addr add fd00:0:0:7::3/64 dev eth 0

Router 1:

- ip -6 addr add fd00:0:0:7::/64 dev eth 1
- ip -6 addr add 2001:1200:0:21f7::/64 dev eth1
- ip -6 addr add fd00::4:/127 dev eth0
- ip -6 addr add 2001:1200:0:21f0::4/127 dev eth0

Servidor Ventas:

- ip -6 addr add fd00:0:0:2::1/64 dev eth 0
- ip -6 addr add 2001:1200:0:21f2::1/64 dev eth0

Pc-Vt-1:

- ip -6 addr add fd00:0:0:2::2/64 dev eth 0
- ip -6 addr add 2001:1200:0:21f2::2/64 dev eth0

Router 7:

- ip -6 addr add fd00:0:0:2::/64 dev eth 0
- ip -6 addr add fd00::11/127 dev eth1
- ip -6 addr add 2001:1200:0:21f2::/64 dev eth0
- ip -6 addr add 2001:1200:0:21f0::11/127 dev eth1

Pc-Vt-2:

- ip -6 addr add fd00:0:0:3::1/64 dev eth 0

Router 8:

- ip -6 addr add fd00:0:0:3::/64 dev eth 1
- ip -6 addr add 2001:1200:0:21f0::13/127 dev eth0
- ip -6 addr add 2001:1200:0:21f3::/64 dev eth1
- ip -6 addr add fd00::13/127 dev eth0

Router 5:

- ip -6 addr add fd00:0:0:4::/64 dev eth 3
- ip -6 addr add fd00::e/127 dev eth0
- ip -6 addr add 2001:1200:0:21f0::e/127 dev eth0
- ip -6 addr add fd00::a/127 dev eth2
- ip -6 addr add fd00::7/127 dev eth1
- ip -6 addr add 2001:1200:0:21f0::7/127 dev eth1
- ip -6 addr add 2001:1200:0:21f4::0/64 dev eth3
- ip -6 addr add 2001:1200:0:21f0::a/127 dev eth2

Router 9:

- ip -6 addr add fd00:0:0:6::/64 dev eth 3
- ip -6 addr add 2001:1200:0:21f6::0/64 dev eth3
- ip -6 addr add fd00:0:0:5::/64 dev eth 2
- ip -6 addr add 2001:1200:0:21f5::0/64 dev eth2
- ip -6 addr add 2001:1200:0:21f0::b/127 dev eth0

- ip -6 addr add 2001:1200:0:21f0::d/127 dev eth1
- ip -6 addr add fd00::b/127 dev eth0
- ip -6 addr add fd00::d/127 dev eth1

Pc-Adm-2:

- ip -6 addr add fd00:0:0:6::2/64 dev eth 0

Servidor Adm:

- ip -6 addr add fd00:0:0:6::1/64 dev eth 0

Pc-Adm-1:

- ip -6 addr add fd00:0:0:5::1/64 dev eth 0

Pc-Casa:

- ip -6 addr add fd00:0:0:1::1/64 dev eth 0
- ip -6 addr add 2001:1200:0:21f1::1/64 dev eth0

Router Casa:

- ip -6 addr add fd00:0:0:1::/64 dev eth 1
- ip -6 addr add 2001:1200:0:21f1::/64 dev eth1

Implementación de Enrutamiento estático

Todos los routers fueron configurados con rutas estáticas para permitir la conectividad interna completa y con la PC-Casa externa.

Pc-Casa:

- ip -6 route add default via 2001:1200:0:21f1:: dev eth 0

Router Casa:

- ip -6 route add default via 192.168.1.2 dev eth 0

ISP Casa:

- ip -6 route add default via 192.168.2.2 dev eth 0

ISP Intranet:

- ip -6 route add default via 192.168.2.1 dev eth 1

Router 2:

- ip -6 route add default via 192.168.3.1 dev eth 2
- ip -6 route add 2001:1200:0:21f0::d/127 via 2001:1200:0:21f0::3 dev eth 1
- ip -6 route add 2001:1200:0:21f0::/60 via 2001:1200:0:21f0::1 dev eth 0
- ip -6 route add fd00:0:0:5::/64 via 2001:1200:0:21f0::3 dev eth 1
- ip -6 route add fd00:0:0:6::/64 via 2001:1200:0:21f0::3 dev eth 1

Router 3:

- ip -6 route add default via 2001:1200:0:21f0::2 dev eth 1
- ip -6 route add 2001:1200:0:21f0::/60 via 2001:1200:0:21f0::9 dev eth 0
- ip -6 route add fd00:0:0:5::/64 via 2001:1200:0:21f0::d dev eth 2
- ip -6 route add fd00:0:0:6::/64 via 2001:1200:0:21f0::d dev eth 2

Router 9:

- ip -6 route add default via 2001:1200:0:21f0::9 dev eth 0
- ip -6 route add 2001:1200:0:21f7::/64 via 2001:1200:0:21f0::c dev eth 1

Pc-Adm-1:

- ip -6 route add default via fd00:0:0:5:: dev eth 0

Servidor Adm:

- ip -6 route add default via fd00:0:0:6:: dev eth 0

Pc-Adm-2:

- ip -6 route add default via fd00:0:0:6:: dev eth 0

Router 5:

- ip -6 route add default via 2001:1200:0:21f0::6 dev eth 1
- ip -6 route add 2001:1200:0:21f0::11/127 via 2001:1200:0:21f0::e dev eth 0
- ip -6 route add 2001:1200:0:21f0::13/127 via 2001:1200:0:21f0::f dev eth 0
- ip -6 route add 2001:1200:0:21f0::11/127 via 2001:1200:0:21f0::f dev eth 0
- ip -6 route add 2001:1200:0:21f2::/64 via 2001:1200:0:21f0::f dev eth 0
- ip -6 route add 2001:1200:0:21f3::/64 via 2001:1200:0:21f0::f dev eth 0
- ip -6 route add 2001:1200:0:21f0::b/127 via 2001:1200:0:21f0::6 dev eth 1
- ip -6 route add 2001:1200:0:21f7::1/64 via 2001:1200:0:21f0::b dev eth 2
- ip -6 route add 2001:1200:0:21f7::2/64 via 2001:1200:0:21f0::b dev eth 2
- ip -6 route add fd00::f/127 via fd00::e dev eth 0
- ip -6 route add fd00::13/127 via 2001:1200:0:21f0::e dev eth 0
- ip -6 route add fd00::11/127 via 2001:1200:0:21f0::e dev eth 0
- ip -6 route add fd00::13/127 via 2001:1200:0:21f0::f dev eth 0
- ip -6 route add fd00::11/127 via 2001:1200:0:21f0::f dev eth 0
- ip -6 route add fd00:0:0:3::/64 via 2001:1200:0:21f0::f dev eth 0
- ip -6 route add fd00:0:0:2::/64 via 2001:1200:0:21f0::f dev eth 0

Pc-Guest:

- ip -6 route add default via 2001:1200:0:21f4:: dev eth 0

Router 6:

- ip -6 route add default via 2001:1200:0:21f0::e dev eth 2
- ip -6 route add 2001:1200:0:21f2::/64 via 2001:1200:0:21f0::11 dev eth 1
- ip -6 route add 2001:1200:0:21f3::/64 via 2001:1200:0:21f0::13 dev eth 0
- ip -6 route add fd00:0:0:3::/64 via 2001:1200:0:21f0::13 dev eth 0
- ip -6 route add fd00:0:0:2::/64 via 2001:1200:0:21f0::11 dev eth 1

Router 8:

- ip -6 route add default via 2001:1200:0:21f0::12 dev eth 0

Pc-Vt-2:

- ip -6 route add default via 2001:1200:0:21f3:: dev eth 0

Router 7:

- ip -6 route add default via 2001:1200:0:21f0::10 dev eth 1

Pc-Vt-1:

- ip -6 route add default via 2001:1200:0:21f2:: dev eth 0

Servidor Ventas:

- ip -6 route add default via 2001:1200:0:21f2:: dev eth 0

Router 4:

- ip -6 route add default via 2001:1200:0:21f0:: dev eth 3
- ip -6 route add 2001:1200:0:21f2::/64 via 2001:1200:0:21f0::7 dev eth 0
- ip -6 route add 2001:1200:0:21f3::/64 via 2001:1200:0:21f0::7 dev eth 0
- ip -6 route add 2001:1200:0:21f4::/64 via 2001:1200:0:21f0::7 dev eth 0
- ip -6 route add 2001:1200:0:21f7::/64 via 2001:1200:0:21f0::4 dev eth 1
- ip -6 route add 2001:1200:0:21f5::/64 via 2001:1200:0:21f0::8 dev eth 2
- ip -6 route add 2001:1200:0:21f6::/64 via 2001:1200:0:21f0::8 dev eth 2
- ip -6 route add fd00::/
- ip -6 route add fd00:0:0:2::/64 via 2001:1200:0:21f0::7 dev eth 0
- ip -6 route add fd00:0:0:3::/64 via 2001:1200:0:21f0::7 dev eth 0

- ip -6 route add fd00:0:0:4::/64 via 2001:1200:0:21f0::7 dev eth 0
- ip -6 route add fd00:0:0:7::/64 via 2001:1200:0:21f0::4 dev eth 1
- ip -6 route add fd00:0:0:5::/64 via 2001:1200:0:21f0::8 dev eth 2
- ip -6 route add fd00:0:0:6::/64 via 2001:1200:0:21f0::8 dev eth 2

Router 1:

- ip -6 route add default via 2001:1200:0:21f0::5 dev eth 0

Pc-Dc-1:

- ip -6 route add default via 2001:1200:0:21f7:: dev eth 0

Servidor DNS:

- ip -6 route add default via 2001:1200:0:21f7:: dev eth 0

Servidor WEB:

- ip -6 route add default via 2001:1200:0:21f7:: dev eth 0

Enrutamiento Basado en Políticas

Se implementaron dos políticas de tráfico específicas:

1. Tráfico desde W-Guest (Red de Invitados)

Los paquetes originados en la red de invitados (2001:1200:0:21f4::1/64) fueron forzados a pasar por Router-9 antes de salir de la intranet, aislando el tráfico de invitados de los recursos internos, excepto los servidores DNS y Web (accesibles mediante sus direcciones globales).

Implementación en Router-5:

- ip -6 route add default via 2001:1200:0:21f0::6 dev eth1
- ip -6 route add 2001:1200:0:21f7::1/64 via 2001:1200:0:21f0::b dev eth2 table 2
- ip -6 route add 2001:1200:0:21f7::2/64 via 2001:1200:0:21f0::b dev eth2 table 2
- ip -6 route add unreachable fd00::/8 table 2
- ip -6 route add unreachable 2001:1200:0:21f0::/60 table 2
- ip -6 rule add from 2001:1200:0:21f4::1/64 table 2

2. Tráfico TCP con destino fuera de la intranet

Los paquetes TCP con destino fuera de la intranet fueron forzados a pasar por Router-3, mientras que el resto del tráfico (UDP, ICMP, etc.) utilizó la ruta por defecto normal.

Implementación en Router-4:

- iptables -t mangle -A PREROUTING -d 2001:1200:0:21f0::/60 -p tcp -j MARK -set mark 2
- ip rule fwmark 2 table 3
- ip -6 route add table 3 default via 2001:1200:0:21f0::8 dev eth2

Implementación en Router-9:

- iptables -t mangle -A PREROUTING -d 2001:1200:0:21f0::/60 -p tcp -j MARK -set mark 2
- ip rule fwmark 2 table 3
- ip -6 route add table 3 default via 2001:1200:0:21f0::c dev eth1

Túneles IPv6 sobre IPv4 (SIT)

Para habilitar la comunicación IPv6 entre la red empresarial y la red hogareña, se estableció un túnel SIT a través de Internet (segmento solo IPv4).

Router 2:

```
ip tunnel add tunelsite mode sit local 192.168.3.2 remote 192.168.1.1
ip link set tunelsite up
ip -6 route add default dev tunelsite
```

Router Casa:

```
ip tunnel add tunnel mode sit local 192.168.1.1 remote 192.168.3.2
ip link set tunnel up
ip -6 route add default dev tunnel
```

La primer línea crea el túnel de tipo site y establece los extremos locales y remotos, La segunda activa el túnel, y la tercera establece la ruta default ipv6 por el túnel creado.

Router Advertisement (RADVD)

Para simplificar la configuración de los clientes, se habilitó RADVD en los routers para anunciar automáticamente los prefijos IPv6. Los clientes obtuvieron direcciones globales y ULA sin intervención manual.

Ejemplo – Router-1 (Data Center):

```
interface eth1 {
    AdvSendAdvert on;
    MinRtrAdvInterval 3;
    MaxRtrAdvInterval 10;
    AdvDefaultPreference low;
    prefix 2001:1200:0:21f7::/64 {
        AdvOnLink on;
        AdvAutonomous on;
        AdvRouterAddr on;
    };
    prefix fd00:0:0:7::/64 {
        AdvOnLink on;
        AdvAutonomous on;
        AdvRouterAddr on;
    };
};
```

Esta configuración se implementó en los routers conectados a redes no punto a punto (Routers 1, 5, 7, 8, 9). Todas las direcciones previamente asignadas manualmente en los clientes fueron eliminadas y reemplazadas por direcciones obtenidas automáticamente mediante SLAAC.

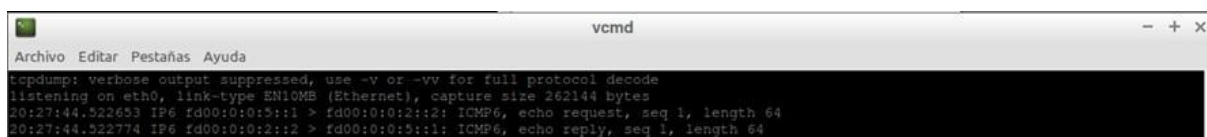
Pruebas de conectividad y análisis de tráfico

Ping de PC-ADM-1 a PC-VT-1

Esta prueba valida la conectividad entre el departamento de Administración y el departamento de Ventas, que se encuentran en diferentes segmentos de red

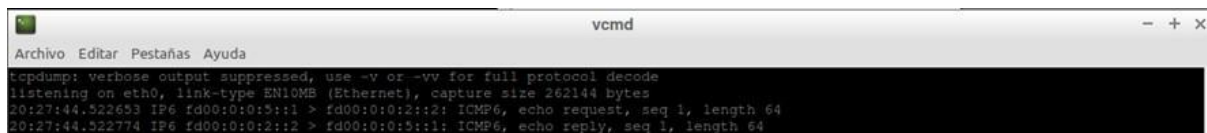
Comando desde PC-ADM-1 : ping6 fd00:0:0:2::2

PC-ADM-1:



```
vcmd
Archivo Editar Pestañas Ayuda
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
20:27:44.522693 IP6 fd00:0:0:5::1 > fd00:0:0:2::2: ICMP6, echo request, seq 1, length 64
20:27:44.522774 IP6 fd00:0:0:2::2 > fd00:0:0:5::1: ICMP6, echo reply, seq 1, length 64
```

PC-VT-1:



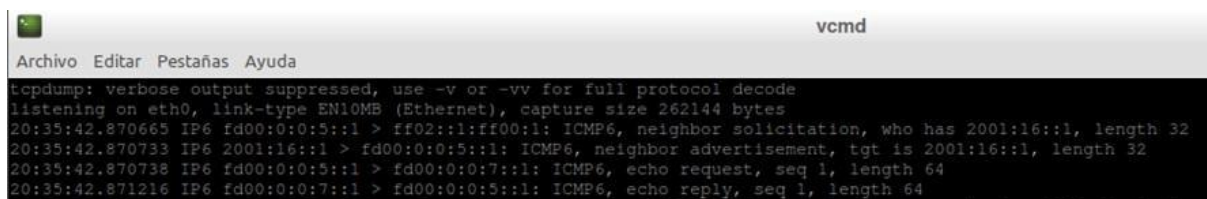
```
vcmd
Archivo Editar Pestañas Ayuda
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
20:27:44.522693 IP6 fd00:0:0:5::1 > fd00:0:0:2::2: ICMP6, echo request, seq 1, length 64
20:27:44.522774 IP6 fd00:0:0:2::2 > fd00:0:0:5::1: ICMP6, echo reply, seq 1, length 64
```

Ping de PC-ADM-1 a SERVIDOR-WEB

Esta prueba verifica que los equipos de Administración puedan acceder al servidor web del Data Center.

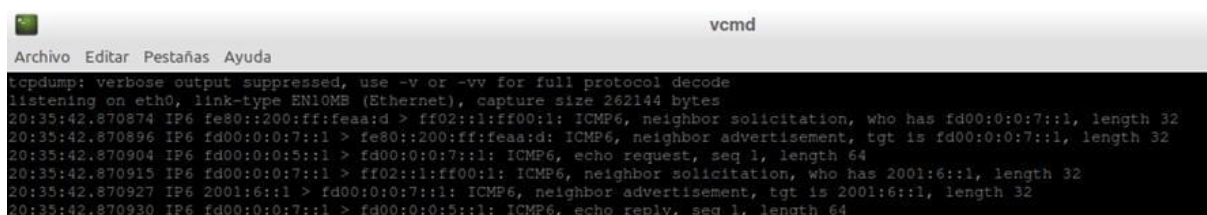
Comando desde PC-ADM-1 : ping6 fd00:0:0:7::1

PC-ADM-1:



```
vcmd
Archivo Editar Pestañas Ayuda
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
20:35:42.870665 IP6 fd00:0:0:5::1 > ff02::1:ff00:1: ICMP6, neighbor solicitation, who has 2001:16::1, length 32
20:35:42.870733 IP6 2001:16::1 > fd00:0:0:5::1: ICMP6, neighbor advertisement, tgt is 2001:16::1, length 32
20:35:42.870738 IP6 fd00:0:0:5::1 > fd00:0:0:7::1: ICMP6, echo request, seq 1, length 64
20:35:42.871216 IP6 fd00:0:0:7::1 > fd00:0:0:5::1: ICMP6, echo reply, seq 1, length 64
```

Servidor-Web:



```
vcmd
Archivo Editar Pestañas Ayuda
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
20:35:42.870874 IP6 fe80::200:ff:feaa:d > ff02::1:ff00:1: ICMP6, neighbor solicitation, who has fd00:0:0:7::1, length 32
20:35:42.870896 IP6 fd00:0:0:7::1 > fe80::200:ff:feaa:d: ICMP6, neighbor advertisement, tgt is fd00:0:0:7::1, length 32
20:35:42.870904 IP6 fd00:0:0:5::1 > fd00:0:0:7::1: ICMP6, echo request, seq 1, length 64
20:35:42.870915 IP6 fd00:0:0:7::1 > ff02::1:ff00:1: ICMP6, neighbor solicitation, who has 2001:6::1, length 32
20:35:42.870927 IP6 2001:6::1 > fd00:0:0:7::1: ICMP6, neighbor advertisement, tgt is 2001:6::1, length 32
20:35:42.870930 IP6 fd00:0:0:7::1 > fd00:0:0:5::1: ICMP6, echo reply, seq 1, length 64
```

Ping de PC-GUEST a SERVIDOR-DNS

Se verificó el cumplimiento de la política de seguridad que restringe el acceso desde la red de invitados. El servidor DNS sólo es accesible desde la red de invitados mediante su dirección global.

Comando desde PC-GUEST : ping6 2001:1200:0:21f7::2

PC-GUEST:

```
vcmd
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tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
20:38:44.033154 IP6 fe80::200:ff:feaa:2a > ff02::1:ff00:1: ICMP6, neighbor solicitation, who has 2001:11::1, length 32
20:38:44.033223 IP6 2001:11::1 > fe80::200:ff:feaa:2a: ICMP6, neighbor advertisement, tgt is 2001:11::1, length 32
20:38:44.033754 IP6 2001:1200:0:21f4::1 > ff02::1:ff00:1: ICMP6, neighbor solicitation, who has 2001:11::1, length 32
20:38:44.033773 IP6 2001:11::1 > 2001:1200:0:21f4::1: ICMP6, neighbor advertisement, tgt is 2001:11::1, length 32
20:38:44.033777 IP6 2001:1200:0:21f4::1 > 2001:1200:0:21f7::2: ICMP6, echo request, seq 1, length 64
20:38:44.034006 IP6 2001:1200:0:21f7::2 > 2001:1200:0:21f4::1: ICMP6, echo reply, seq 1, length 64
```

Servidor-Dns:

```
vcmd
Archivo Editar Pestañas Ayuda
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
20:38:44.033906 IP6 fe80::1200:ff:feaa:d > ff02::1:ff00:2: ICMP6, neighbor solicitation, who has 2001:1200:0:21f7::2, length 32
20:38:44.033924 IP6 2001:1200:0:21f7::2 > fe80::1200:ff:feaa:d: ICMP6, neighbor advertisement, tgt is 2001:1200:0:21f7::2, length 32
20:38:44.033933 IP6 2001:1200:0:21f4::1 > 2001:1200:0:21f7::2: ICMP6, echo request, seq 1, length 64
20:38:44.033945 IP6 2001:1200:0:21f7::2 > ff02::1:ff00:1: ICMP6, neighbor solicitation, who has 2001:6::1, length 32
20:38:44.033957 IP6 2001:6::1 > 2001:1200:0:21f7::2: ICMP6, neighbor advertisement, tgt is 2001:6::1, length 32
20:38:44.033959 IP6 2001:1200:0:21f7::2 > 2001:1200:0:21f4::1: ICMP6, echo reply, seq 1, length 64
```

Descubrimiento de vecinos IPv6

Se realizó una prueba de multicast para verificar el funcionamiento del protocolo de descubrimiento de vecinos en el Data Center.

Comando desde Router-1: ping6 ff02::1%eth1

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	fe80::200:ff:feaa:d	ff02::1	ICMPv6	118	Echo (ping) request id=0x0036, seq=1, hop limit=1 (multicast)
2	0.000068180	fe80::db:94ff:fe7a::	ff02::1:ffaa:d	ICMPv6	86	Neighbor Solicitation for fe80::200:ff:feaa:d from 12:76:74:b4:d0:7f
3	0.000126562	fe80::200:ff:feaa:10	ff02::1:ffaa:d	ICMPv6	86	Neighbor Solicitation for fe80::200:ff:feaa:d from 00:00:00:aa:00:10
4	0.000130858	fe80::200:ff:feaa:f	fe80::200:ff:feaa:d	ICMPv6	118	Echo (ping) reply id=0x0036, seq=1, hop limit=64
5	0.000132538	fe80::200:ff:feaa:e	ff02::1:ffaa:d	ICMPv6	86	Neighbor Solicitation for fe80::200:ff:feaa:d from 00:00:00:aa:00:0e
6	0.000121822	fe80::200:ff:feaa:d	fe80::db:94ff:fe7a::	ICMPv6	86	Neighbor Advertisement fe80::200:ff:feaa:d (rtr, sol, ovr) is at 00:00:00:aa:00:0d
7	0.000141676	fe80::db:94ff:fe7a::	fe80::200:ff:feaa:d	ICMPv6	118	Echo (ping) reply id=0x0036, seq=1, hop limit=64
8	0.000149444	fe80::200:ff:feaa:d	fe80::200:ff:feaa:10	ICMPv6	86	Neighbor Advertisement fe80::200:ff:feaa:d (rtr, sol, ovr) is at 00:00:00:aa:00:0d
9	0.000156928	fe80::200:ff:feaa:d	fe80::200:ff:feaa:e	ICMPv6	86	Neighbor Advertisement fe80::200:ff:feaa:d (rtr, sol, ovr) is at 00:00:00:aa:00:0d
10	0.000168713	fe80::200:ff:feaa:10	fe80::200:ff:feaa:d	ICMPv6	118	Echo (ping) reply id=0x0036, seq=1, hop limit=64
11	0.000169668	fe80::200:ff:feaa:e	fe80::200:ff:feaa:d	ICMPv6	118	Echo (ping) reply id=0x0036, seq=1, hop limit=64
12	1.005501893	fe80::200:ff:feaa:d	ff02::1	ICMPv6	118	Echo (ping) request id=0x0036, seq=2, hop limit=1 (multicast)
13	1.005560040	fe80::db:94ff:fe7a::	fe80::200:ff:feaa:d	ICMPv6	118	Echo (ping) reply id=0x0036, seq=2, hop limit=64
14	1.005591508	fe80::200:ff:feaa:10	fe80::200:ff:feaa:d	ICMPv6	118	Echo (ping) reply id=0x0036, seq=2, hop limit=64
15	1.005592750	fe80::200:ff:feaa:f	fe80::200:ff:feaa:d	ICMPv6	118	Echo (ping) reply id=0x0036, seq=2, hop limit=64
16	1.005593526	fe80::200:ff:feaa:e	fe80::200:ff:feaa:d	ICMPv6	118	Echo (ping) reply id=0x0036, seq=2, hop limit=64
17	2.031557149	fe80::200:ff:feaa:d	ff02::1	ICMPv6	118	Echo (ping) request id=0x0036, seq=3, hop limit=1 (multicast)
18	2.031645152	fe80::db:94ff:fe7a::	fe80::200:ff:feaa:d	ICMPv6	118	Echo (ping) reply id=0x0036, seq=3, hop limit=64
19	2.031695016	fe80::200:ff:feaa:10	fe80::200:ff:feaa:d	ICMPv6	118	Echo (ping) reply id=0x0036, seq=3, hop limit=64
20	2.031696667	fe80::200:ff:feaa:f	fe80::200:ff:feaa:d	ICMPv6	118	Echo (ping) reply id=0x0036, seq=3, hop limit=64
21	2.031697623	fe80::200:ff:feaa:e	fe80::200:ff:feaa:d	ICMPv6	118	Echo (ping) reply id=0x0036, seq=3, hop limit=64

El comando ping6 ff02::1%eth1 envía un mensaje ICMPv6 Echo Request a la dirección multicast ff02::1, que alcanza a todos los nodos IPv6 en el enlace local. Todos los equipos en el Data Center respondieron con sus direcciones link-local, confirmando:

- Correcto funcionamiento del stack IPv6
- Configuración automática de direcciones link-local
- Multicast operativo en la red

Netcat entre servidor de ventas y PC-Casa

Para validar la conectividad a nivel de aplicación, se estableció una comunicación UDP entre el Servidor de Ventas y la PC-Casa a través del túnel SIT.

SERVIDOR-VENTAS (escucha):

vcmd

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```
root@SERVIDOR-VENTAS:/tmp/pycore.45523/SERVIDOR-VENTAS.conf# nc -6 -l -u 2020
hola
```

Apply a display filter ... <Ctrl-F>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	2001:1200:0:21f1::1	2001:1200:0:21f2::1	UDP	67	43889 → 2020 Len=5
2	5.246941660	fe80::200:ff:feaa:2	2001:1200:0:21f2::1	ICMPv6	86	Neighbor Solicitation for 2001:1200:0:21f2::1 from 00:00:00:aa:00:02
3	5.246974212	2001:1200:0:21f2::1	fe80::200:ff:feaa:2	ICMPv6	78	Neighbor Advertisement 2001:1200:0:21f2::1 (sol)
4	10.366629792	fe80::200:ff:feaa:1	fe80::200:ff:feaa:2	ICMPv6	86	Neighbor Solicitation for fe80::200:ff:feaa:2 from 00:00:00:aa:00:01

Frame 1: 67 bytes on wire (536 bits), 67 bytes captured (536 bits) on interface 0

Ethernet II, Src: 00:00:00:aa:00:02 (00:00:00:aa:00:02), Dst: 00:00:00:aa:00:01 (00:00:00:aa:00:01)

Destination: 00:00:00:aa:00:01 (00:00:00:aa:00:01)

Source: 00:00:00:aa:00:02 (00:00:00:aa:00:02)

Type: IPv6 (0x86dd)

Internet Protocol Version 6, Src: 2001:1200:0:21f1::1, Dst: 2001:1200:0:21f2::1

0110 = Version: 6

.... 0000 0000 = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)

.... 1000 1000 0110 0001 0000 = Flow Label: 0x88610

Payload Length: 13

Next Header: UDP (17)

Hop Limit: 58

Source: 2001:1200:0:21f1::1

Destination: 2001:1200:0:21f2::1

User Datagram Protocol, Src Port: 43889, Dst Port: 2020

Source Port: 43889

Destination Port: 2020

Length: 13

Checksum: 0xc5c6 [unverified]

[Checksum Status: Unverified]

[Stream index: 0]

Data (5 bytes)

Data: 686f6c610a

[Length: 5]

PC- Casa (envío):

vcmd

Archivo Editar Pestañas Ayuda

```
<523/PC-Casa.conf# nc -6 -u 2001:1200:0:21f2::1 2020
hola
```

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	fe80::200:ff:feaa:1e	fe80::200:ff:feaa:1d	ICMPv6	86	Neighbor Solicitation for fe80::200:ff:feaa:1d from 00:00:00:aa:00:1e
2	0.000042281	fe80::200:ff:feaa:1d	fe80::200:ff:feaa:1e	ICMPv6	78	Neighbor Advertisement fe80::200:ff:feaa:1d (rtr, sol)
3	24.256617328	2001:1200:0:21f1::1	2001:1200:0:21f2::1	UDP	67	43889 → 2020 Len=5
4	29.439912128	fe80::200:ff:feaa:1e	2001:13::1	ICMPv6	86	Neighbor Solicitation for 2001:13::1 from 00:00:00:aa:00:1e
5	29.440482224	2001:13::1	fe80::200:ff:feaa:1e	ICMPv6	78	Neighbor Advertisement 2001:13::1 (rtr, sol)
6	34.560074147	fe80::200:ff:feaa:1d	fe80::200:ff:feaa:1e	ICMPv6	86	Neighbor Solicitation for fe80::200:ff:feaa:1e from 00:00:00:aa:00:1d
7	34.560106091	fe80::200:ff:feaa:1e	fe80::200:ff:feaa:1d	ICMPv6	78	Neighbor Advertisement fe80::200:ff:feaa:1e (sol)
8	39.681170152	fe80::200:ff:feaa:1e	fe80::200:ff:feaa:1d	ICMPv6	86	Neighbor Solicitation for fe80::200:ff:feaa:1d from 00:00:00:aa:00:1e
9	39.681252409	fe80::200:ff:feaa:1d	fe80::200:ff:feaa:1e	ICMPv6	78	Neighbor Advertisement fe80::200:ff:feaa:1d (rtr, sol)
10	63.104979921	2001:1200:0:21f1::1	2001:1200:0:21f2::1	UDP	67	43889 → 2020 Len=5
11	68.351936910	fe80::200:ff:feaa:1e	2001:13::1	ICMPv6	86	Neighbor Solicitation for 2001:13::1 from 00:00:00:aa:00:1e

Frame 3: 67 bytes on wire (536 bits), 67 bytes captured (536 bits) on interface 0

Ethernet II, Src: 00:00:00:aa:00:1e (00:00:00:aa:00:1e), Dst: 00:00:00:aa:00:1d (00:00:00:aa:00:1d)

Destination: 00:00:00:aa:00:1d (00:00:00:aa:00:1d)

Source: 00:00:00:aa:00:1e (00:00:00:aa:00:1e)

Type: IPv6 (0x86dd)

Internet Protocol Version 6, Src: 2001:1200:0:21f1::1, Dst: 2001:1200:0:21f2::1

0110 = Version: 6

.... 0000 0000 = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)

.... 1000 1000 0110 0001 0000 = Flow Label: 0x88610

Payload Length: 13

Next Header: UDP (17)

Hop Limit: 64

Source: 2001:1200:0:21f1::1

Destination: 2001:1200:0:21f2::1

User Datagram Protocol, Src Port: 43889, Dst Port: 2020

Source Port: 43889

Destination Port: 2020

Length: 13

Checksum: 0xc5c6 [unverified]

[Checksum Status: Unverified]

[Stream index: 0]

Data (5 bytes)

Data: 686f6c610a

[Length: 5]

En el nivel Ethernet se ven las MAC de origen y destino junto con la versión IP que se está utilizando (ipv6).

En el nivel de internet hay información del paquete como la longitud, el protocolo de transporte (UDP), el Hop Limit y las direcciones Ipv6 del origen y destino.

En el nivel de transporte (UDP), se tienen los puertos de destino y origen, la longitud del header y el checksum.

Por último se ven los datos que se envían y su longitud.

Conectividad Externa y Túneles SIT

Se verificó la conectividad entre la red hogareña (PC-Casa) y los servidores de la empresa a través del túnel IPv6 sobre IPv4.

Ping de PC-Casa a SERVIDOR-DNS: ping6 2001:1200:0:21f7::2

ROUTER- Casa eth1 (casa):

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	2001:1200:0:21f1::1	2001:1200:0:21f2::2	ICMPv6	118	Echo (ping) request id=0x003a, seq=1, hop limit=64 (reply in 2)
2	0.000137296	2001:1200:0:21f2::2	2001:1200:0:21f1::1	ICMPv6	118	Echo (ping) reply id=0x003a, seq=1, hop limit=58 (request in 1)
3	5.198019724	fe80::200:ff:feaa:1d	2001:1200:0:21f1::1	ICMPv6	86	Neighbor Solicitation for 2001:1200:0:21f1::1 from 00:00:00:aa:00:1d
4	5.198303972	2001:1200:0:21f1::1	fe80::200:ff:feaa:1d	ICMPv6	78	Neighbor Advertisement 2001:1200:0:21f1::1 (sol)
5	107.853962138	fe80::c2:dcff:febe...	ff02::2	ICMPv6	70	Router Solicitation from 02:c2:dc:be:69:e1
6	140.676486403	fe80::8a9c:a9ff:feh...	ff02::2	ICMPv6	70	Router Solicitation from 02:c2:dc:be:69:e1
▶ Frame 1: 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface 0						
▶ Ethernet II, Src: 00:00:00:aa:00:1e (00:00:00:aa:00:1e), Dst: 00:00:00:aa:00:1d (00:00:00:aa:00:1d)						
▼ Internet Protocol Version 6, Src: 2001:1200:0:21f1::1, Dst: 2001:1200:0:21f2::2						
0110 = Version: 6						
▶ 0000 0000 = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)						
▶ 1101 1110 0001 0100 1010 = Flow Label: 0xde14a						
Payload Length: 64						
Next Header: ICMPv6 (58)						
Hop Limit: 64						
Source: 2001:1200:0:21f1::1						
Destination: 2001:1200:0:21f2::2						
▶ Internet Control Message Protocol v6						

ROUTER- Casa eth0 (internet):

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	2001:1200:0:21f1::1	2001:1200:0:21f2::2	ICMPv6	138	Echo (ping) request id=0x003a, seq=1, hop limit=63 (reply in 2)
2	0.000111722	2001:1200:0:21f2::2	2001:1200:0:21f1::1	ICMPv6	138	Echo (ping) reply id=0x003a, seq=1, hop limit=59 (request in 1)
3	5.198182861	00:00:00:aa:00:1b	00:00:00:aa:00:1c	ARP	42	Who has 192.168.1.1? Tell 192.168.1.2
4	5.198094208	00:00:00:aa:00:1c	00:00:00:aa:00:1b	ARP	42	Who has 192.168.1.2? Tell 192.168.1.1
5	5.198228138	00:00:00:aa:00:1c	00:00:00:aa:00:1b	ARP	42	192.168.1.1 is at 00:00:00:aa:00:1c
6	5.198292389	00:00:00:aa:00:1b	00:00:00:aa:00:1c	ARP	42	192.168.1.2 is at 00:00:00:aa:00:1b
7	173.390158855	fe80::b0c5:6fff:fe5...	ff02::2	ICMPv6	70	Router Solicitation from b2:c5:6f:59:cf:de
▶ Frame 1: 138 bytes on wire (1104 bits), 138 bytes captured (1104 bits) on interface 0						
▶ Ethernet II, Src: 00:00:00:aa:00:1c (00:00:00:aa:00:1c), Dst: 00:00:00:aa:00:1b (00:00:00:aa:00:1b)						
▼ Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.3.2						
0100 = Version: 4						
.... 0101 = Header Length: 20 bytes (5)						
▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)						
Total Length: 124						
Identification: 0xaa5f (43615)						
Flags: 0x4000, Don't fragment						
Time to live: 63						
Protocol: IPv6 (41)						
Header checksum: 0x0ba6 [validation disabled]						
[Header checksum status: Unverified]						
Source: 192.168.1.1						
Destination: 192.168.3.2						
▼ Internet Protocol Version 6, Src: 2001:1200:0:21f1::1, Dst: 2001:1200:0:21f2::2						
0110 = Version: 6						
▶ 0000 0000 = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)						
▶ 1101 1110 0001 0100 1010 = Flow Label: 0xde14a						
Payload Length: 64						
Next Header: ICMPv6 (58)						
Hop Limit: 63						
Source: 2001:1200:0:21f1::1						
Destination: 2001:1200:0:21f2::2						
▶ Internet Control Message Protocol v6						

Se observa un paquete ICMPv6 (Echo Request) originado en 2001:1200:0:21f1::1 (PC-Casa) con destino a 2001:1200:0:21f2::2 (PC-VT-1), el cual viaja encapsulado dentro de un datagrama IPv4.

La cabecera IPv4 externa muestra:

- Origen: 192.168.1.1 (Router-Casa)
- Destino: 192.168.3.2 (Router-2)
- Protocolo: 41, que indica que el payload transporta un datagrama IPv6 completo.

Esta encapsulación permite que el tráfico IPv6 atraviese la red del ISP (que solo soporta IPv4), demostrando que el túnel SIT está operativo y que el tráfico interno de la empresa y la red hogareña pueden comunicarse exitosamente.

Traceroute a través del Túnel.

Se realizó traceroute desde PC-Casa hacia el Servidor DNS para verificar la ruta que toman los paquetes.

PC- Casa:

```
root@PC-Casa:/tmp/pycore.39163/PC-Casa.conf# traceroute6 2001:1200:0:21f7::2
traceroute to 2001:1200:0:21f7::2 (2001:1200:0:21f7::2) from 2001:1200:0:21f1::1, 30 hops max, 24 byte packets
 1 2001:1200:0:21f1:: (2001:1200:0:21f1::)  0,059 ms  0,024 ms  0,017 ms
 2  * * *
 3  * * *
 4 2001:1200:0:21f0::4 (2001:1200:0:21f0::4)  0,16 ms  0,041 ms  0,033 ms
 5 2001:1200:0:21f7::2 (2001:1200:0:21f7::2)  0,046 ms  0,036 ms  0,032 ms
```

ROUTER- Casa eth0 (internet)

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	2001:1200:0:21f1::1	2001:1200:0:21f7::2	UDP	106	48416 → 33434 Len=24
2	0.000018341	192.168.1.2	192.168.1.1	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
3	5.002166614	2001:1200:0:21f1::1	2001:1200:0:21f7::2	UDP	106	48416 → 33434 Len=24
4	5.002192413	192.168.1.2	192.168.1.1	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
5	5.088834102	00:00:00:aa:00:1c	00:00:00:aa:00:1b	ARP	42	Who has 192.168.1.2? Tell 192.168.1.1
6	5.088963693	00:00:00:aa:00:1b	00:00:00:aa:00:1c	ARP	42	192.168.1.2 is at 00:00:00:aa:00:1b
7	10.005938275	2001:1200:0:21f1::1	2001:1200:0:21f7::2	UDP	106	48416 → 33434 Len=24
▶ Frame 1: 106 bytes on wire (848 bits), 106 bytes captured (848 bits) on interface 0						
▶ Ethernet II, Src: 00:00:00:aa:00:1c (00:00:00:aa:00:1c), Dst: 00:00:00:aa:00:1b (00:00:00:aa:00:1b)						
▼ Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.3.2						
0100 = Version: 4						
.... 0101 = Header Length: 20 bytes (5)						
▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)						
Total Length: 92						
Identification: 0x80d7 (32983)						
▶ Flags: 0x4000, Don't fragment						
▶ Time to live: 1						
Protocol: IPv6 (41)						
Header checksum: 0x734e [validation disabled]						
[Header checksum status: Unverified]						
Source: 192.168.1.1						
Destination: 192.168.3.2						
▼ Internet Protocol Version 6, Src: 2001:1200:0:21f1::1, Dst: 2001:1200:0:21f7::2						
0110 = Version: 6						
▶ 0000 0000 = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)						
.... 1111 1001 1111 1010 0111 = Flow Label: 0x79fa7						
Payload Length: 32						
Next Header: UDP (17)						
Hop Limit: 1						
Source: 2001:1200:0:21f1::1						
Destination: 2001:1200:0:21f7::2						
▶ User Datagram Protocol, Src Port: 48416, Dst Port: 33434						
▶ Data (24 bytes)						

La captura en la interfaz eth0 del Router-Casa muestra los paquetes UDP enviados por traceroute, encapsulados dentro de IPv4 a través del túnel SIT.